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ENERGETIC COST OF LOCOMOTION IN ANIMALS

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Abstract—1. The minimum cost of transport (the minimum ratio of metabolic rate to the product of speed and body weight) for various walking, running and flying animals is functionally related to the type of transport and to body weight. The same functional relations hold for poikilotherms and homeotherms, and for five or more decades of body weight.

2. The cost of transport for a flying animal is less than that for a walker or a runner of the same size. The two costs differ by a factor of 27 for animals weighing 0·01 kg.

3. Large birds and mammals have costs of transport that are about the same as those for automobiles and small propeller airplanes.

MOST ANIMALS have some means of moving themselves from one place to another at the expense of metabolic energy. It is interesting to compare the effectiveness of the various devices that have evolved to accomplish locomotion. Before this can be done, one must choose some measure of effectiveness that can be applied with useful results to animals whose sizes may differ by orders of magntitude, and whose locomotor processes may be as different as walking and flying. The measure of effectiveness that I have chosen expresses the energetic cost of transporting a unit of body weight for a unit of distance. This "cost of transport" is the ratio of power input to the product of speed of transport and body weight, where the power input of an animal is its metabolic rate.

Cost of transport with these dimensions has been used by Weis-Fogh (1952) to compare flight in insects and man-made aircraft, and by Gabrielli & von Karman (1950) to compare transport in a large variety of man-made vehicles. The latter authors provide a valuable discussion of cost of transport as a measure of the effectiveness of locomotion. However, since they use a different definition of power than that given above, their numerical values cannot be compared with those in this study simply by converting to common units.

The cost of transport has convenient numerical values for animals and intuitive meaning when it is expressed as kcal/(kg km). However, it actually is a dimensionless number and is related to the efficiency of the transport process (Gabrielli & von Karman, 1950). Efficiency (e) is the ratio of the power output (P_o) of a device to power input (P_i), and I have already mentioned that the power input of an animal is its metabolic rate. Power output may be defined as the rate at which work is done on the environment through which the animal is moving at velocity V.

Work is done to move against a force (F) which is due to aerodynamic or hydrodynamic drag and a component of weight if the velocity vector V is not level. The force vector F is parallel to V but in the opposite direction. Thus, power output is given by

$$P_o = FV. \tag{1}$$

Since

$$P_o/P_i = e, \tag{2}$$

$$P_i = FV/e, \tag{3}$$

and

$$\text{cost of transport} = P_i/(VW) = F/(eW), \tag{4}$$

where W is body weight. The cost of transport is dimensionless since e is dimensionless and the force dimensions of F and W cancel in equation (4). Equation (4) also shows that the cost of transport is directly proportional to the resistance of the environment and inversely proportional to the efficiency of the transport process.

The cost of transport is related to several of the familiar measures of economy used for man-made vehicles. For a particular vehicle, it is directly proportional to liters of fuel consumed per 100 km, the measure of automobile economy in Europe, and inversely proportional to the miles per gallon measure used in the United States. In addition, if the vehicle is using fuel with a constant caloric content per unit weight, the cost of transport is proportional to the fractional weight of the vehicle that is used as fuel to cover a given distance. Thus, when the cost of transport is least, the maximum distance can be covered with the available fuel.

Since the cost of transport is inversely proportional to velocity, it varies from infinity when the animal is at rest to some minimum value when speed is finite. In this paper, I have tabulated minimum values for costs of transport for various animals and vehicles moving on the level from data available in the literature. Some of these data have been published previously without documentation (Tucker, 1969).

The minimum cost of transport for different animals depends on body weight and type of locomotion (Fig. 1). Values for walking and running animals ranging in size from a mouse to a horse decrease as body weight increases according to the equation

$$\frac{\text{kcal}}{\text{kg km}} = 0{\cdot}100(10)^E, \tag{5}$$

where

$$E = 1{\cdot}67W^{-0{\cdot}126} \tag{6}$$

and W is in kg. The data for walkers and runners can be linearized by plotting log $\{\log[\text{kcal}/(\text{kg km})] + 1\}$ against log W. Equations (5) and (6) were obtained from a linear least-squares regression line (transformed to Fig. 1) fitted to such a plot. The regression line had a standard error of estimate of 0·0446 log $\{\log[\text{kcal}/(\text{kg km})] + 1\}$ for a sample size of 20.

The equations show that cost of transport changes proportionally more with size in smaller animals than in larger animals. Thus, the cost of transport for a 0·1 kg animal is 18 per cent that of a 0·01 kg animal, but the cost of transport for a 1000 kg animal is 58 per cent that of a 100 kg animal.

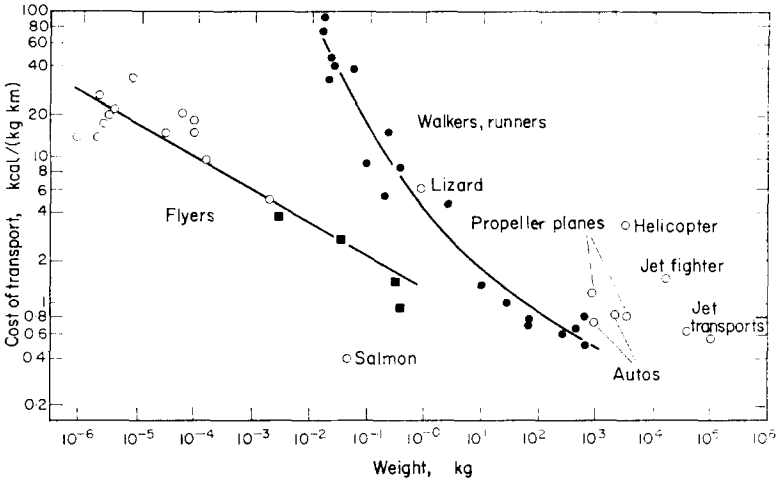

FIG. 1. Minimum costs of transport for insects (unlabelled open circles), birds (squares), mammals (filled circles) and other animals and man-made vehicles (labelled open circles) with various weights and types of locomotion. Cost of transport can be converted to a dimensionless number by multiplying by the factor 0·427 kg km/kcal. Alternatively, it can be converted to percentage of body weight used as fuel/km by multiplying by 0·011 kg %/kcal if the fuel is fat, 0·088 kg %/kcal if the fuel is glycogen and 0·0098 kg %/kcal if the fuel is gasoline. The factor for glycogen includes the 2·7 kg of water that is stored with each kg of glycogen (Weis-Fogh, 1952). See Table 1 for data and references.

Values for minimum cost of transport for flying animals ranging in size from a fruit fly to a pigeon fall around a straight line on double-logarithmic co-ordinates. The linear least-squares relation for such a plot has the equation

$$\frac{\text{kcal}}{\text{kg km}} = 1{\cdot}25W^{-0{\cdot}227} \tag{7}$$

with a standard error of estimate of 0·186 log [kcal/(kg km)] for a sample size of 16. The cost of transport decreases as size increases so that the cost of transport for a particular animal is 59 per cent of that for another animal weighing a tenth as much.

There is a striking difference in cost of transport between walkers and runners and fliers. The latter are much more economical. For example, a bird weighing 0·01 kg uses only one-twenty-seventh as much energy to cover a given distance as a mouse of the same weight. Similarly, a swarm of 3000 bee-sized insects each weighing 10^{-4} kg could fly for a given distance for about the same total energy

cost as a single 0·3 kg rat-sized mammal could run it. It is no wonder that small flying animals undertake long seasonal migrations but small terrestrial animals do not.

Poikilotherms seems to have about the same costs of transport for their weight as homeotherms, whether walking or flying. Flying poikilotherms are represented in Fig. 1 by insects, walking poikilotherms by a lizard.

The lowest cost of transport yet measured is that for a young salmon weighing 0·048 kg. To swim a given distance at a temperature of 5°C, it uses only 1·5 to 17 per cent as much energy as a running mammal or a flying bird, respectively, of the same weight.

For comparison with animals, I have included some data in Fig. 1 for various vehicles. The larger birds are about as economical as small propeller planes or jet fighters, and are better than helicopters. A jet transport is more economical than any bird yet measured, but extrapolation of the line for fliers indicates that a swan or goose weighing 10 kg might do as well. The larger walking and running mammals do about as well as jet transports or automobiles.

TABLE 1—DATA AND REFERENCES FOR MINIMUM COST OF TRANSPORT

Animal or vehicle	Wt (kg)	Speed (km/hr)	Metabolic rate [kcal/(kg hr)]	Reference
FLYERS:				
Fruit fly (*Drosophila*)	$2{\cdot}00 \times 10^{-6}$	7·40	104	Vogel, 1965; Weis-Fogh, 1952
	$9{\cdot}00 \times 10^{-7}$	4·70	65·0	Hocking, 1953
Black-fly (*Simulium*)	$2{\cdot}50 \times 10^{-6}$	4·30	73·5	Hocking, 1953
	$2{\cdot}10 \times 10^{-6}$	8·80	234	Hocking, 1953
Mosquito (*Aedes*)	$3{\cdot}20 \times 10^{-6}$	2·50	47·2	Hocking, 1953
	$8{\cdot}20 \times 10^{-6}$	3·20	112	Hocking, 1953
Blow fly (*Lucilia*)	$3{\cdot}00 \times 10^{-5}$	30·0	450	Weis-Fogh, 1952
Honey bee (*Apis*)	$1{\cdot}00 \times 10^{-4}$	30·0	450	Weis-Fogh, 1952
	$1{\cdot}00 \times 10^{-4}$	7·90	142	Hocking, 1953
Horse fly (*Tabanus*)	$6{\cdot}20 \times 10^{-5}$	9·10	177	Hocking, 1953
	$1{\cdot}62 \times 10^{-4}$	7·20	69·3	Hocking, 1953
Desert locust (*Schistocerca*)	$2{\cdot}00 \times 10^{-3}$	15·0	75·0	Weis-Fogh, 1952
Hummingbird (*Calypte*)	$3{\cdot}00 \times 10^{-3}$	49·0*	197	Lawsieski, 1963
Budgerigar (*Melopsittacus*)	0·0350	35·0	102	Tucker, 1968
Laughing Gull (*Larus*)	0·310	31·0	44·9	Tucker, 1969
Pigeon (*Columba*)	0·384	58·0	55·1	LeFebvre, 1964

* Speed assumed, metabolic rate measured while hovering.

TABLE 1—*cont.*

Animal or vehicle	Wt (kg)	Speed (km/hr)	Metabolic rate (kcal/[kg hr)]	Reference
WALKERS AND RUNNERS:				
Mouse (*Microtus*)	0·0180	0·540	50·7	Jansky, 1959b
Mouse (*Clethrionomys*)	0·0180	0·370	27·1	Jansky, 1959b
Mouse (*Peromyscus*)	0·0230	1·08	37·8	Hart & Heroux, 1954
Mouse (*Mus*)	0·0240	0·560	27·4	Jansky, 1959a
Mouse (*Mus*)	0·0270	0·670	28·6	Hart, 1950
Lemming (*Dicrostonyx*)	0·0610	1·54	63·0	Hart & Heroux, 1955
Kangaroo rat (*Dipodomys*)	0·100	2·10	19·3	Taylor & Schmidt-Nielsen, 1969, and personal communication
Ground squirrel (*Citellus*)	0·198	2·80	15·2	Taylor & Schmidt-Nielsen, 1969, and personal communication
Rat (*Rattus*)	0·397	2·10	18·0	Taylor & Schmidt-Nielsen, 1969, and personal communication
Rat (*Rattus*)	0·250	1·08	16·2	Hart & Jansky, 1963
Lizard (*Iguana*)	0·900	0·370	2·31	Moberly, 1968
Rabbit	2·54	2·10	10·0	Hart & Heroux, 1955
Dog	10·7	5·84	7·94	Dill, 1963
Sheep	30·0	2·88	2·88	Blaxter, 1962; Clapperton, 1961
Man	70·0	5·62	4·12	Henry, 1953
Man	70·0	5·58	4·06	Dill, 1965
Pony	270	0·354	2·22	Brody, 1945
Colt	477	3·54	2·43	Brody, 1945
Cattle	656	3·54	2·90	Brody, 1945
Horse	707	5·00	2·61	Brody, 1945
VEHICLES: †				
Cherokee airplane	978	200	240	Stanfield, 1967
Volkswagen automobile	1010	81·2	59·6	D'Angelo, 1968
Cadillac automobile	2210	119	98·3	D'Angelo, 1968
Sikorsky S62 helicopter	3590	148	524	Stanfield, 1967
Grand Commander airplane	3860	371	308	Stanfield, 1967
F105F jet fighter	17,200	1150	1750	Taylor, 1968
DC 9–10 jet transport	41,300	879	566	Stanfield, 1967
DC 8 jet transport	107,000	965	550	Taylor, 1968

† Caloric equivalent of fuel between 10,200 and 10,400 kcal/kg.

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